Ayush Priyadarshi

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Research Interests

Computational Physics • Scientific Software Development • Monte Carlo Simulations • Geant4 • Detector-Response Modelling • Radiation Detection • Gamma Spectroscopy • High-Performance Computing • Scientific Instrumentation

Education

Indian Institute of Technology Roorkee **November 2021 – Expected April/May 2027**
Integrated Bachelor of Science–Master of Science in Physics Roorkee, Uttarakhand, India

Current CGPA: 4.318/10

Relevant Coursework: Programming, Computational Physics, Nuclear Physics, Optics, Electronics, Quantum Mechanics, Condensed Matter Physics, Laboratory Physics

Thesis: *A Python-Assisted Geant4 Workflow for Automated Scintillation Detector Response Simulations*
Supervisor: Dr. Anil Kumar Gourishetty **Thesis Performance:** B+ in Thesis Stage I; A in Thesis Stage II

Research Experience

Undergraduate Thesis Research **2025–2026**
A Python-Assisted Geant4 Workflow for Automated Scintillation Detector Response Simulations
Department of Physics, Indian Institute of Technology Roorkee
Supervisor: Dr. Anil Kumar Gourishetty

Developed an automated Geant4–Python workflow for campaign-level scintillation detector response simulations by integrating a Geant4 C++ backend with Python-based macro generation, simulation execution, ROOT validation, spectral analysis, count extraction, plotting, and table generation.

- Implemented and maintained a configurable Geant4 simulation backend for scintillation detector studies.
- Developed Python automation scripts for macro generation, batch execution, ROOT-file validation, spectral analysis, and count-table generation.
- Supported monoenergetic gamma simulations, radioactive-decay simulations, beta-plus decay cases, material-dependent studies, detector-size studies, and geometry-dependent studies.
- Implemented solid cylindrical, hollow annular, and custom near-4π soccer-ball detector geometries.
- Performed detector-response studies for NaI, LaBr3, CsI, BGO, GGAG, and PbWO4 scintillators.
- Processed ROOT outputs to generate deposited-energy spectra and extract photopeak, sum-peak, and total non-zero energy-deposition counts.
- Executed high-statistics Geant4 simulations on the PARAMGanga HPC facility, including 10^8-event validation runs.
- Consolidated the workflow into a public GitHub repository used for ongoing detector simulation research.

Publications

A. Priyadarshi, G. A. Kumar, K. Madhan, and V. Ranga, "A Python-based Tool for Automated Geant4 Simulations of Scintillation Detectors," submitted to *IEEE Nuclear Science Symposium*, 2026.
Status: Under review.

Research Software

Geant4 Detector Simulation Pipeline **Developer and Maintainer**
github.com/ayushpriyadarshi-18/geant4-detector-simulation-pipeline

Python-assisted Geant4 simulation pipeline for automated scintillation detector response studies. The software integrates Geant4 C++ detector simulations with Python tools for macro generation, campaign execution, ROOT-file validation, spectrum generation, peak-count extraction, total-count extraction, plotting, and output organisation.

- Supports automated monoenergetic gamma and radioactive-decay simulations.
- Includes solid cylindrical, hollow annular, and near-4π soccer-ball detector geometries.
- Supports NaI, LaBr3, CsI, BGO, GGAG, and PbWO4 scintillator materials.
- Provides ROOT-based deposited-energy analysis using `EdepCrystal_keV`.
- Generates reproducible spectra, count tables, logs, and reference outputs.
- Includes public Cs-137 reference demonstrations across multiple geometries.
- Used as the software basis for a first-author IEEE NSS 2026 submission.

Technical Skills

Programming: Python, C++, SQL, Bash, Fortran, HTML/CSS
Scientific Software: Geant4, ROOT, Monte Carlo simulations, detector-response modelling
Research Automation: Macro generation, campaign execution, ROOT validation, spectral analysis, plotting, CSV/table generation
Tools: Git, Linux, Docker, Qt, MySQL
HPC: PARAMGanga, batch simulation workflows, high-statistics Geant4 simulations
Embedded Systems: Raspberry Pi, Arduino, ESP8266, microcontrollers, FPGA, relay circuits
Systems/Networking: Linux administration, Docker services, Samba/NFS, reverse proxy, DNS filtering, network gateway configuration

Laboratory Skills

Radiation Detection: NaI(Tl) and LaBr3 scintillation detectors, gamma spectroscopy, radioactive sources, detector signal acquisition
Nuclear Electronics: NIM/VME modules, pulse processing, counting systems, oscilloscopes
Instrumentation: FPGA-based systems, microcontrollers, Arduino, ESP8266, Raspberry Pi, relay circuits
Optics: Lasers, optical alignment, interference, diffraction, spectrometers, optical benches
General Physics Labs: Condensed matter, atmospheric physics, thermal physics, electronics, optics, and nuclear physics laboratories

Engineering and Instrumentation Projects

Radio Telescope Control and Embedded Linux Support
Physics and Astronomy Club, IIT Roorkee **Role: Assisting Contributor**

Contributed to Raspberry Pi-based embedded Linux and hardware-control support for a student-led radio telescope project.

- Assisted in developing a custom Raspberry Pi image for the radio telescope system.
- Modified boot configuration for reliable startup under lower-voltage portable power conditions.
- Enabled networking over the power/data port for a single-cable portable Linux setup.
- Built support for an initially unsupported Wi-Fi module.
- Assisted with hardware-control integration for telescope operation.

Raspberry Pi Home Server and Network Gateway
Independent systems engineering project

Designed and maintain a Raspberry Pi-based home server for self-hosted services, network management, and automation.

- Deployed Docker-based services including Jellyfin, Home Assistant, Samba/NFS file sharing, reverse proxy services, DNS filtering, and network gateway functionality.
- Gained practical experience in Linux administration, networking, Docker, service deployment, and home automation.

WiFi IP Mouse
First Prize, Syntax Error Hackathon, SDS Labs, IIT Roorkee, 2022
github.com/ayushpriyadarshi-18/MyPi

Built a WiFi-based IP mouse system using Raspberry Pi during the Syntax Error Hackathon.

- Developed a Raspberry Pi-based wireless mouse/control system.
- Implemented network-based communication between the Raspberry Pi and host device.
- Built the project under hackathon time constraints.

IoT Smart Switch with Alexa Integration
Independent embedded systems project, 2020

Built a Wi-Fi-enabled smart switch using an ESP8266 microcontroller and relay-based switching to control household lights and fans through Alexa voice commands.

Awards

- **Category Winner**, Solar Decathlon India Net-Zero Building Competition – Rs. 1,00,000 cash prize.
- **First Prize**, Syntax Error Hackathon, SDS Labs, IIT Roorkee, 2022 – Built a Raspberry Pi-based WiFi IP Mouse.

References

References available upon request.